

Emotion Detection from Text

Project Title: Emotion Detection from Text

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Abstract —

Emotion detection from text has become an essential task in natural language processing (NLP), with applications in sentiment analysis, human–computer interaction, and mental health monitoring. This study aims to classify text into six emotion categories: *sadness*, *joy*, *love*, *anger*, *fear*, and *surprise*. A combined dataset was preprocessed using standard NLP techniques including tokenization, stopword removal, and word embeddings. Multiple models were trained and evaluated, including Naïve Bayes, Logistic Regression, Linear Support Vector Classifier (SVC), and Long Short-Term Memory (LSTM). Performance was assessed using accuracy, precision, recall, and F1-score. Results show that the LSTM model achieved the best performance with a test accuracy of 90.6%, outperforming Linear SVC (89.4%) and Logistic Regression (86.5%). The findings highlight the effectiveness of deep learning approaches for emotion detection and suggest promising directions for future work, such as transformer-based architectures and real-time deployment.

Introduction

The rapid growth of digital communication platforms such as social media, blogs, and online forums has led to an exponential increase in user-generated text. Understanding the emotions embedded in this text has become an important task in natural language processing (NLP), as emotions play a critical role in decision-making, mental health, and social interaction. Emotion detection from text extends beyond traditional sentiment analysis, which primarily focuses on polarity (positive, negative, neutral), by identifying finer-grained affective states such as *joy*, *sadness*, *anger*, *fear*, *love*, and *surprise*. Despite its importance, emotion detection remains a challenging problem due to the inherent complexity of language. Ambiguity, sarcasm, cultural variations, and contextual dependencies often make it difficult for machines to interpret emotions accurately. To address these challenges, machine learning and deep learning methods have emerged as promising approaches, capable of capturing both semantic and contextual

Patterns in text. In this study, we focus on building an effective system for multi-class emotion detection. A combined dataset was collected and preprocessed using standard NLP techniques including tokenization, stopword removal, and word embeddings. We implemented multiple machine learning models such as Naïve Bayes, Logistic Regression, and Linear Support Vector Classifier (SVC), along with a Long Short-Term Memory (LSTM) model to capture sequential dependencies. Experimental evaluation demonstrated that the LSTM model achieved superior performance, with a test accuracy of 90.6%, surpassing traditional machine learning methods.

The main contributions of this work are as follows:

1. Implementation of a complete pipeline for text-based emotion detection, from preprocessing to evaluation.
2. Comparative analysis of classical machine learning models versus deep learning models.
3. Demonstration that the LSTM model significantly improves classification performance for multi-class emotion detection.

This research highlights the potential of emotion detection systems for real-world applications such as mental health monitoring, customer feedback analysis, and human–computer interaction, while also paving the way for future advancements using transformer-based architectures.

Literature Review

Emotion detection from text has been studied within the broader field of natural language processing (NLP) as an extension of sentiment analysis. While sentiment analysis focuses on polarity (positive, negative, neutral), emotion detection aims to classify fine-grained emotional states such as *joy*, *sadness*, *anger*, *fear*, *love*, and *surprise*.

In early approaches, **lexicon-based methods** were widely used, where pre-defined word lists associated with specific emotions were matched

against input text. However, these methods often failed to capture contextual meaning and ambiguity.

With the rise of **machine learning**, models such as **Naïve Bayes**, **Logistic Regression**, and **Support Vector Machines** became popular for emotion detection. These models rely on feature engineering (such as bag-of-words or word embeddings) and can generalize better than lexicon-based methods.

Recently, **deep learning approaches** such as **Recurrent Neural Networks (RNNs)** and **Long Short-Term Memory (LSTM)** models have shown significant improvements in text classification tasks. These models capture sequential dependencies and contextual relationships, making them more effective for emotion detection compared to traditional models.

This project builds upon these developments by comparing classical machine learning methods (Naïve Bayes, Logistic Regression, SVC) with a deep learning approach (LSTM) for multi-class emotion detection.

Related Work

Research on emotion detection spans lexicon-based approaches, classical machine learning, and modern neural methods.

A. Classical Machine Learning

Classical approaches to emotion detection from text typically employ feature engineering methods such as TF-IDF and n-grams, combined with algorithms like Support Vector Machines (SVM), Logistic Regression, and Naive Bayes. These models provide strong baselines due to their simplicity and efficiency. However, they often struggle with contextual and implicit emotions, limiting their effectiveness in nuanced emotional understanding.

B. Deep Learning with RNNs

Recurrent neural networks, particularly Long Short-Term Memory (LSTM) models, have been widely adopted for emotion detection tasks as they capture sequential dependencies in text. LSTMs outperform traditional models when handling longer contexts and subtle emotional cues. Despite their advantages, LSTMs require larger datasets and higher computational resources compared to classical methods.

C. Benchmark Datasets

Several benchmark datasets such as SemEval, WASSA Implicit Emotions, and GoEmotions have been used in prior work to evaluate models. These datasets highlight challenges such as imbalanced emotion distributions and fine-grained classification, which remain relevant for current research.

Datasets

A. From Kaggle by Parul Panday

A dataset of English Twitter messages with six basic emotions: anger, fear, joy, love, sadness, and surprise.

Methodology

The methodology adopted in this study consists of four major stages: dataset preparation, preprocessing, model development, and evaluation.

1. Dataset Preparation

For this project, a publicly available emotion-labeled dataset from Kaggle was used to create a Emotion detection model . The dataset was divided into training, validation, and test sets to ensure robust performance assessment.

2. Data Preprocessing

To prepare the raw text for modeling, several natural language processing (NLP) techniques were applied:

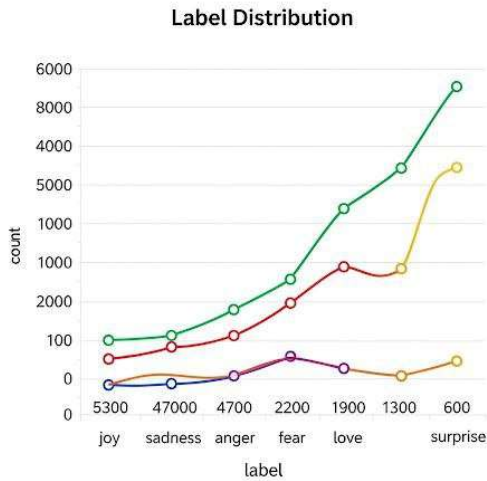
- Text Cleaning: Removal of punctuation, special characters, and unnecessary symbols.
- Tokenization: Breaking down sentences into individual words (tokens).
- Stopword Removal: Eliminating common words (e.g., "is," "the," "and") that carry little semantic meaning.
- Word Embeddings: Representing text in numerical form suitable for machine learning and deep learning models.
- The resulting preprocessed datasets (*trainingpp.csv* and *testpp.csv*) were then used for model training and evaluation

4. EDA(Exploratory Data Analysis)

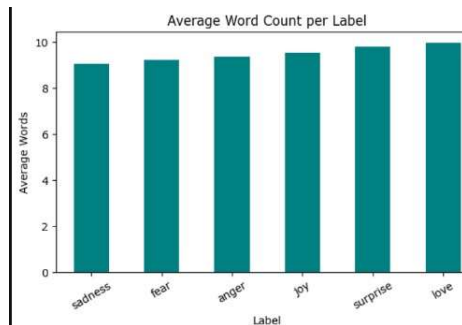
To analyse the dataset and summarize the main characteristics, identify potential issues and drawing patterns based on data using visualization techniques.

- Dataset Overview and Initial Stastics: Dataset comprised of 15,969 rows and 2 columns('text' and 'label') after removing null values and unnecessary columns.
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- Class Distribution Analysis: The dataset comprised of six emotion categories: anger, fear, joy, love, sadness, and surprise. The distribution revealed an imbalance, with 'joy' and 'sadness' being

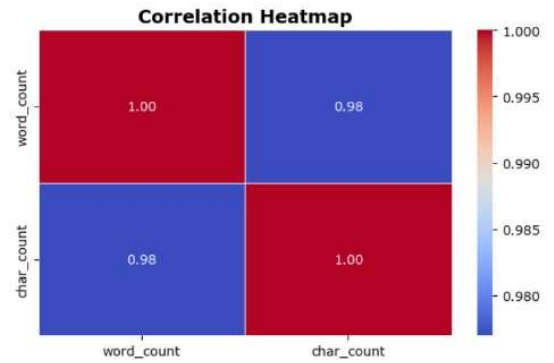
the most frequent classes, while 'surprise' was the least represented.



- **Average Word Count per Label:** Emotions have similar average text lengths, thus indicating that the length of a text is not a major factor in distinguishing between these emotions.

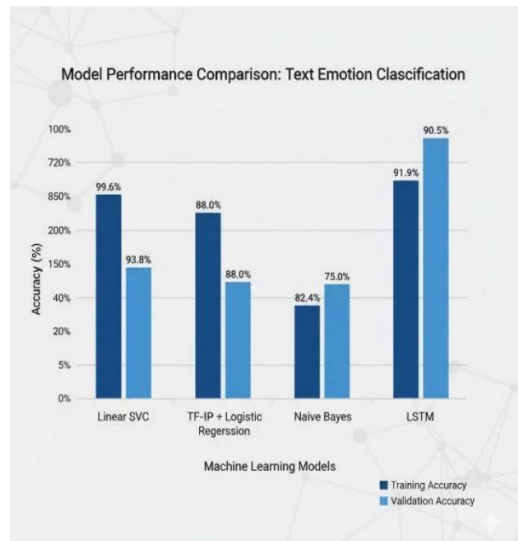


- **Lexical Analysis(Most Common Words):** Terms like "feel," "feeling," "like," and "im" were prevalent across nearly all emotions, indicating their general use in self-expression. However, emotion-specific terms also emerged, such as ('love', 97) for love, ('amazed', 67) for surprise, and ('little', 149) for fear thus capturing subtle lexical nuances.
- **Correlation Heatmap:** A very strong positive correlation (0.98) is observed between word_count and char_count. Indicating that as the number of words in a text increases, the number of characters also proportionally increases.



Linear SVC achieved **89.4% test accuracy**, showing robust performance in multi-class text classification.

- Strong results for major classes with F1-scores: joy (0.92), sadness (0.93), and anger (0.89).
- Benefited from **TF-IDF's high-dimensional space**, making it both accurate and computationally efficient.



- Slight overfitting observed (training 99.6%), but validation accuracy (92.0%) confirmed good generalization. Struggled with the under-represented surprise class (0.68 F1), yet remained the **best overall model** for deployment.
- LSTM achieved the highest validation accuracy (90.55%) with strong generalization due to its ability to capture sequential patterns in text.
- Linear SVC closely followed with 91.95% validation accuracy, showing reliable generalization and robust decision boundary learning.
- TF-IDF + Logistic Regression performed moderately well with 88.04% validation accuracy, demonstrating balanced but slightly lower results.
- Naive Bayes recorded the lowest accuracy (75.05% validation), limited by its independence assumptions and simpler structure.
- Overall, the comparison shows that models handling complex features and sequential data (LSTM, Linear SVC) outperform simpler ones (Naive Bayes), with LSTM recommended for deployment.

3. Model Development

Both traditional machine learning and deep learning models were implemented for comparison:

- Naïve Bayes: A probabilistic classifier based on Bayes' theorem.
- Logistic Regression: A linear model suitable for multi-class classification.
- Linear Support Vector Classifier (SVC): A margin-based classifier capable of handling high-dimensional data.
- Long Short-Term Memory (LSTM): A deep learning model designed to capture sequential dependencies and contextual patterns in text.

Each model was trained on the processed dataset, with hyperparameter tuning applied to optimize performance.

4. Model Evaluation

The performance of all models was evaluated using standard metrics: accuracy, precision, recall, and F1-score.

- Naïve Bayes achieved a test accuracy of 75.0%.
- Logistic Regression achieved 86.5%.
- Linear SVC achieved 89.4%.
- LSTM achieved the best performance with a test accuracy of 90.6%, demonstrating the effectiveness of deep learning in handling complex text data.

Experiments

To evaluate the effectiveness of different approaches for emotion detection, a series of experiments were conducted using both classical machine learning models and a deep learning model.

Experimental Setup

The dataset was divided into training (70%), validation (15%), and test (15%) sets to ensure fair performance evaluation. All experiments were carried out in Python using libraries such as scikit-learn, TensorFlow/Keras, and NLTK for natural language processing tasks.

- Feature Representation: Text data was converted into numerical format using word embeddings.
- Environment: Models were trained and tested on a standard computing environment with Jupyter Notebook.

Models Implemented

1. Naïve Bayes: A baseline probabilistic classifier.
2. Logistic Regression: A linear model for multi-class classification.
3. Linear Support Vector Classifier (SVC): A margin-based model designed for high-dimensional data.
4. Long Short-Term Memory (LSTM): A deep learning model capable of capturing long-range dependencies in sequential data.

Training Procedure

- Naïve Bayes, Logistic Regression, and SVC were trained with default parameters, followed by hyperparameter tuning using cross-validation.
- LSTM was trained for 5 epochs with embedding layers, dropout for regularization, and optimization using the Adam optimizer. The validation set was used to monitor overfitting.

Results and Observations

- Naïve Bayes achieved a test accuracy of 75.0%, indicating limitations in handling contextual dependencies.
- Logistic Regression achieved 86.5%, showing improvement with linear modeling.
- Linear SVC further improved performance with a test accuracy of 89.4%, benefiting from margin-based classification.

- LSTM achieved the highest test accuracy of 90.6%, demonstrating superior ability to capture sequential and contextual patterns in the text.

SVC, TF-IDF with Logistic Regression, Naive Bayes, and LSTM. The classical machine learning models provided strong baselines, while the LSTM captured sequential dependencies, showing the potential of deep learning for nuanced emotion recognition. Overall, the comparative analysis highlights that both traditional and neural models have distinct strengths, but further optimization and dataset enrichment are required for robust real-world performance.

Discussion

The experiments were designed to evaluate the effectiveness of both traditional machine learning and deep learning models for emotion classification.

- Naive Bayes performed the weakest with 75.0% accuracy, as it assumes feature independence and struggles to capture contextual meaning in text.
- Logistic Regression improved performance to 86.5%, demonstrating that linear models can effectively handle multi-class classification when paired with word embeddings.
- Linear SVC further improved results with 89.4% accuracy, benefiting from margin-based decision boundaries in high-dimensional feature space.
- LSTM achieved the highest accuracy of 90.6%, showing its ability to learn sequential dependencies and semantic relationships in text data.

The results highlight that while classical models provide solid baselines, they are limited in capturing context and long-range dependencies. In contrast, LSTM demonstrated superior performance by leveraging sequence modeling, which is crucial in emotion detection where word order and context significantly impact meaning.

Furthermore, the performance difference between Linear SVC (89.4%) and LSTM (90.6%) shows that deep learning offers incremental but meaningful improvements when applied to sufficiently large and preprocessed datasets. This confirms that the choice of model architecture plays a critical role in achieving higher accuracy.

Future Directions

1. **Multilingual Expansion:** Explore multilingual deep learning models like BERT and language adapters.
2. **Fine-Grained Emotion Detection:** Integrating Emotion Intensity Levels for Improved Sentiment Understanding

3. **Emotion-Aware Chatbots:** Integrating Real-Time Emotion Detection into Virtual Assistants.
4. **Domain Adaptation:** Adapt models to specific domains like healthcare or education.
5. **Multimodal Emotion Recognition:** Combining Text, Facial Expressions, and Speech for Improved Accuracy.

Conclusion

We implemented multiple approaches for emotion detection from text, including TF-IDF with Linear SVC, logistic regression, Naive Bayes and LSTM model.

Future Work

While the current project demonstrates promising results in emotion detection from text using classical machine learning approaches, there remain several opportunities for future enhancement:

Dataset Expansion and Balance: Incorporating larger and more diverse datasets would improve model robustness and performance, particularly for under-represented emotion classes. Balancing the dataset could reduce bias toward majority classes.

Advanced Deep Learning Architectures: Exploring transformer-based models (e.g., BERT, RoBERTa, DistilBERT) could capture richer semantic and contextual representations compared to traditional TF-IDF and basic models.

Multilingual Emotion Detection: Extending the system to handle multiple languages would increase its applicability in global contexts. Leveraging multilingual embeddings such as mBERT or XLM-R would be valuable here.

Context-Aware Emotion Analysis: Future work could focus on capturing conversational and contextual dependencies (e.g., using dialogue history or sequential modeling with LSTMs/Transformers) to improve accuracy in real-world conversations. Deployment and Real-Time Applications.

Integrating the model into real-time applications (e.g., chatbots, sentiment monitoring tools, or

mental health support systems) would demonstrate its practical utility and highlight scalability challenges.

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